

Shanghai Metro Line Planning: Reproduction and Line-Pool Selection

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Abstract—This paper studies urban rail line planning through both algorithm reproduction and a compact line-selection framework of our own. We first reproduce two representative TNDP methods: the greedy extension and insertion heuristics of Laporte et al. (2005) for single-alignment trip coverage maximization, and the GIS-integrated genetic algorithm of Ahmed et al. (2020) for simultaneous multi-line station and alignment optimization. We then propose Candidate Line-Pool Selection (CLPS), which reduces computation by generating plausible lines in advance and selecting from this finite pool using Greedy-OD, Greedy-BCR, GA-pool, and ILP selectors. All methods are evaluated on a 54-station Shanghai candidate graph derived from OpenStreetMap data. The reproduction results confirm the polynomial runtime scaling predicted by complexity analysis and show that simultaneous multi-line optimization outperforms individual line optimization by 18%. CLPS-Greedy-OD reaches 51.9% OD coverage in only 0.01 s, demonstrating the value of reducing route design to line-pool selection. The best OD coverage overall is achieved by Laporte H2-C (56.8%), while multi-line methods provide better transfer structure and operational flexibility.

Index Terms—transit network design, line-pool selection, greedy heuristics, genetic algorithm, OD demand capture, Shanghai metro

I. INTRODUCTION

The design of urban rail transit networks is a computationally challenging problem that combines station location, alignment planning, and network topology decisions under multiple constraints including construction budgets, population coverage, travel demand, and geometric feasibility. Formally known as the Transit Network Design Problem (TNDP), it is NP-hard in the general case, motivating the development of practical heuristics.

This paper has two goals. The first is reproduction: we implement two influential algorithms from the TNDP literature that represent complementary approaches to the problem. Laporte et al. (2005) formulate the single-alignment problem as maximizing origin-destination (OD) trip coverage under a length budget, and propose simple greedy heuristics that run in polynomial time. Ahmed et al. (2020) extend the scope to multi-line networks by integrating GIS-based feasibility screening with a genetic algorithm (GA) that simultaneously optimizes station locations and line alignments.

The second goal is our own reduced-search planning framework. Directly searching over all ordered station sequences and multi-line combinations is expensive even for a moderate 54-station candidate set. We therefore build Candidate

Line-Pool Selection (CLPS): a two-stage framework that first generates plausible lines on a corridor graph and then selects a small network from this pool. All methods are tested on a realistic dataset derived from OpenStreetMap (OSM) for central Shanghai.

We contribute: (1) a faithful implementation of the Laporte 2005 greedy extension and insertion heuristics with incremental effectiveness computation; (2) an adapted implementation of the Ahmed 2020 GA with hub-based terminal selection and geometric quality constraints; (3) a CLPS framework with Greedy-OD, Greedy-BCR, GA-pool, and ILP selectors; and (4) a unified comparison of reproduction methods and CLPS methods on coverage, runtime, transfer, detour, and passenger-flow metrics.

II. RELATED WORK

A. Overview of TNDP Methods

The Transit Network Design Problem (TNDP) has been studied since the 1970s [7], with four main objective classes: (1) maximizing coverage, (2) minimizing travel time, (3) minimizing cost, and (4) multi-objective combinations. Early work by Dufourd et al. (1996) and Bruno et al. (2002) applied Tabu search to single-line population coverage. Laporte et al. (2005) shifted to OD trip coverage with polynomial greedy heuristics—our first reproduction target. On the multi-line front, Chai et al. (2019) used simulated annealing, while Ahmed et al. (2020) combined GIS feasibility screening with a GA for simultaneous multi-line optimization—our second target. The line-pool approach of Schöbel and Scholl (2005) motivates our CLPS framework: instead of searching directly over all possible station sequences, we generate a manageable pool of plausible candidate lines and solve a smaller line-selection problem.

B. Laporte et al. (2005): Single Alignment Maximization

Given candidate stations S with trip coverage values f_{ij} and a logit mode-choice function $g_{ij}(E)$, the MTCP seeks an ordered subset $E \subseteq S$ maximizing ridership subject to length budget $L_{\max}$:

$$\begin{aligned} \max_{E \subseteq S} \quad & \sum_{i,j \in E} n_{ij}(E) \\ \text{s.t.} \quad & n_{ij}(E) = f_{ij} \cdot g_{ij}(E), \\ & \text{Length}[\text{Align}(E)] \leq L_{\max}. \end{aligned} \tag{1}$$

This generalizes the bounded-length Hamiltonian path problem (NP-hard).

The paper proposes two constructive heuristics: **H1 (Greedy Extension)** starts from the best station pair and repeatedly extends at either endpoint; **H2 (Greedy Insertion)** inserts the best unselected station at its optimal position, with variants A/B/C differing in selection criterion (max ridership, min length, max ratio). A **post-optimization** step reinserts each station at its best position until convergence.

With incremental effectiveness computation, H1 runs in $O(n^3)$ and H2 in $O(n^4)$, where $n = |S|$. Post-optimization costs $O(k^3)$ per pass.

C. Ahmed et al. (2020): Integrated Multi-Line Optimization

Ahmed et al. minimize total system cost:

$$\min T_c = \phi_p P_c + \phi_o O_c + \phi_c C_c \quad (2)$$

where P_c , O_c , C_c are passenger, operator, and community costs. Stage 1 uses GIS to screen feasible station cells; Stage 2 applies a GA with tournament selection, uniform crossover, and station-level mutation. Chromosomes encode entire solutions (lines as genes, stations as bits); terminals are pre-specified. Constraints include 4–7 stations per line, 800–1500 m spacing, and $\leq 30\%$ line overlap. Total complexity is $O(N_{\text{cells}}K + G \cdot P \cdot L \cdot N_s^2 \log N_s)$.

III. DATA, REPRODUCTION, AND CLPS IMPLEMENTATION

A. Data Pipeline

Following the project proposal, we separate spatial data preparation from the route-optimization stage. The optimization algorithms do not operate directly on raw GIS layers. Instead, their common input is a prepared candidate-station set $V = \{1, \dots, n\}$, where each station i has a coordinate $p_i = (\lambda_i, \phi_i)$ and a nonnegative value v_i . Candidate edges are then assigned geographic construction lengths ℓ_{ij} , and the optimization stage searches for ordered station sequences or multi-line subgraphs over this reduced spatial graph.

We instantiate this proposal pipeline using public OpenStreetMap (OSM) data for central Shanghai (121.40–121.56°E, 31.14–31.26°N). First, we extract building footprints and functional OSM features such as commercial, office, retail, university, hospital, transport, and amenity points. Residential population is estimated from building footprint area under a fixed study-area total of 300k residents, while job and activity weights are assigned from the functional features. Second, we form candidate urban centers by running weighted k -means on population, job, and commercial/activity weights, and add three fixed high-importance hubs to ensure that major urban anchors are retained. Nearby centers within 1,000 m are merged, producing a final 54-station candidate set with station values ranging from 30k to 635k.

Third, we convert the station set into a feasible planning graph. Rather than using the complete graph for every algorithm, we build an MST-plus-short-edges corridor graph: the minimum spanning tree preserves global connectivity, while

Candidate Station Selection: Residents + Jobs + Activities

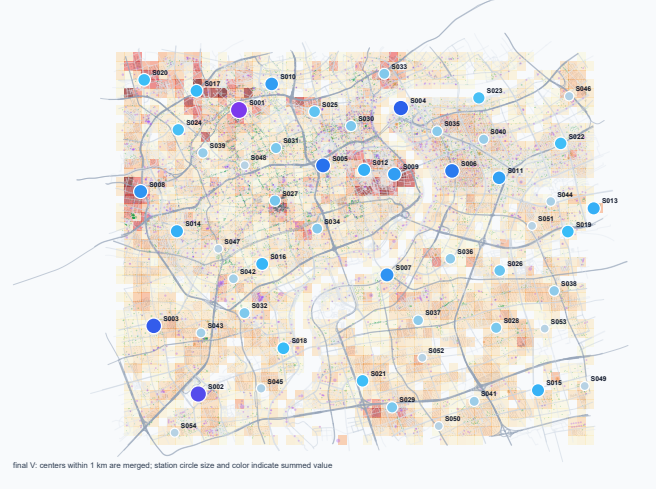


Figure 1. Final 54 candidate stations generated from the Shanghai OSM preprocessing pipeline. Marker size reflects the station value used by the optimization algorithms.

additional short station pairs provide local alignment alternatives. This graph supplies edge lengths, shortest-path distances, and geometric constraints for the CLPS and GA methods. Finally, because real census OD data are unavailable, we generate a gravity-model demand proxy $OD_{ij} \propto \text{pop}_i^{1.5} \cdot \text{jobs}_j / d_{ij}^\gamma$. Thus the final prepared dataset consists of station coordinates, station values, corridor edges, edge lengths, and an OD matrix shared by all reproduced algorithms and CLPS methods.

B. Laporte 2005 Implementation

We implement H1, H2-A/B/C, and post-optimization in 334 lines of Python, using incremental effectiveness computation ($O(|P|)$ gain per station addition). Three simplifications from the original: (1) gravity-model OD replaces census-tract trip coverage (justified by the paper’s statement that heuristics are model-independent); (2) $g_{ij}(E) = 1$ (no logit mode choice); (3) linear distance-decay $E(P) = \sum_{i < j} v_i v_j \max(0, 1 - D_{ij}^P/C)$ to make path ordering meaningful.

C. Ahmed 2020 Implementation

We implement the Stage 2 GA in 829 lines of Python with these adaptations: (1) our station pipeline replaces GIS grid screening (54 vs. 4,774 candidates); (2) data-driven hub-based terminal selection (first line uses highest-OD pair, subsequent lines share a terminal) replaces planner-specified terminals; (3) shortest-path initialization on the corridor graph replaces random sequences; (4) three mutation operators (swap, insert, remove) with feasibility checks; (5) value-connection fitness replaces the paper’s life-cycle cost model; (6) spacing constraints relaxed for our sparse layout.

D. Candidate Line-Pool Selection

In addition to the two reproduced algorithms, we implement a third family of methods called **Candidate Line-Pool Selection (CLPS)**. Its purpose is to reduce computation. Directly

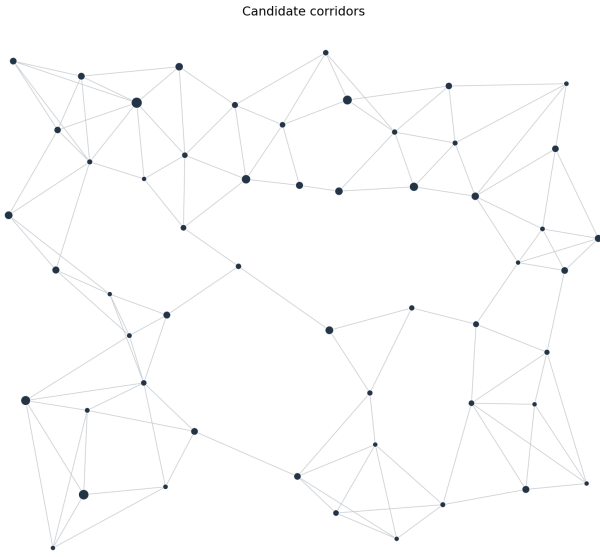
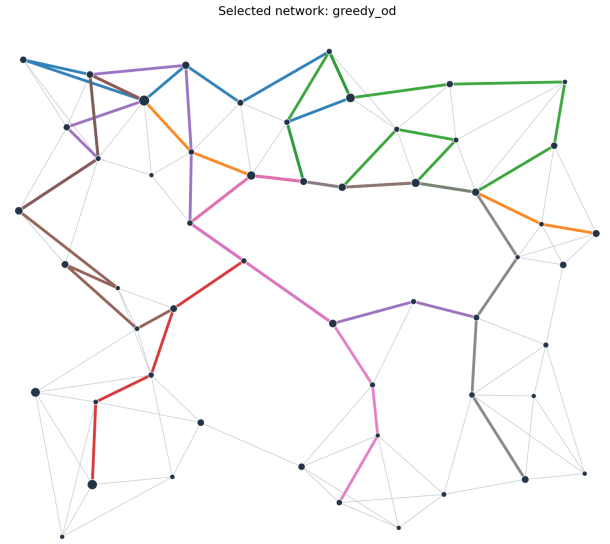


Figure 2. Intermediate CLPS corridor graph used for candidate line-pool generation. Gray edges show the MST-plus-short-edges planning graph from which shortest-path and random-walk candidate lines are sampled.

optimizing metro lines over 54 stations requires searching over a very large space of ordered station sequences and multi-line combinations. CLPS decomposes this into two smaller stages: first generate a finite pool of plausible lines, then select a subset of lines under optional line-count and budget constraints.

The line pool is generated on the MST-plus-short-edges corridor graph. We combine three sources of candidate sequences: (1) shortest paths between the top 120 OD-demand station pairs; (2) shortest paths between pairs among the top 20 high-value terminal candidates; and (3) 200 biased random walks that start from value-weighted stations and prefer short corridor edges. All sequences are filtered to 4–18 stations (random-walk lines use at most 12 stations), reversed duplicates are removed, and each remaining line is annotated with length, construction cost, covered population, and direct OD coverage.

We then solve the reduced line-selection problem with four selectors: **Greedy-OD** repeatedly chooses the line with the largest marginal new direct OD coverage; **Greedy-BCR** chooses the largest marginal coverage per construction cost; **GA-pool** uses a binary chromosome over candidate lines with crossover, mutation, and a fitness combining OD coverage, population, station coverage, connectivity, and cost; and **ILP** uses binary line variables to maximize direct OD coverage with optional maximum-line and budget constraints via CBC. CLPS is used only as our own reduced-search planning framework; the Laporte and Ahmed reproductions do not depend on the line pool.



(a) CLPS-Greedy-OD



(b) CLPS-Greedy-BCR

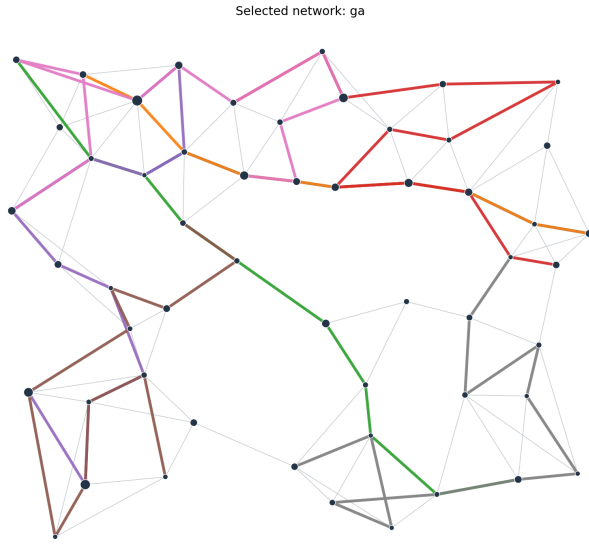
Figure 3. Greedy CLPS selected networks from the same candidate line pool. Greedy-OD maximizes marginal direct OD coverage, while Greedy-BCR favors shorter high-benefit-per-cost lines.

IV. EVALUATION AND CROSS-METHOD COMPARISON

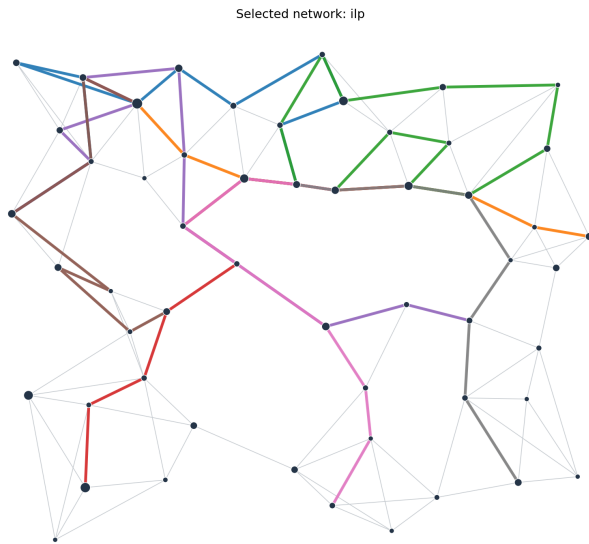
A. Laporte 2005 Reproduction

Table I reproduces the paper’s Table 1 structure on our Shanghai dataset with three length budgets.

Deviation from the paper: The original paper reports H1 dominates H2 on simulated data. Our results show the opposite: H2-A outperforms H1 by 2–2.5 $\times$. This is because without logit mode choice, OD capture depends on the *set* of stations rather than *ordering*; H2 packs more stations (31 vs. 18 at $L_{\max} = 60$ km) by inserting anywhere in the alignment. Post-optimization yields negligible improvement, consistent with the paper.



(a) CLPS-GA-pool



(b) CLPS-ILP

Figure 4. Optimization-based CLPS selected networks. GA-pool explores broader population and connectivity trade-offs, while ILP selects the same line set as Greedy-OD in this run.

B. Runtime Scalability

Figure 5 confirms the predicted scaling: H1 runs in 0.02 s at $n = 20$ and 0.81 s at $n = 100$; H2-A in 0.07 s and 21.2 s respectively. All runtimes are consistent with the paper’s conclusion that “computational times are always insignificant.”

C. Ahmed 2020: Multi-Line Results

Figure 6 shows the optimized 3-line network with hub-based radial topology. Table II summarizes GA results. Simultaneous optimization achieves 18% better cost than individual optimization (Table III), consistent with the paper’s qualitative

Table I
LAPORTE 2005 TABLE 1 REPRODUCTION ON SHANGHAI DATA.

$L_{\max}$	Method	Len. (km)	Ridership	Ratio	T (s)
40	H1	39.9	4.10×10^{12}	1.03×10^8	0.03
40	H2-A	39.6	1.04×10^{13}	2.61×10^8	0.31
40	H2-C	39.2	1.08×10^{13}	2.75×10^8	0.38
60	H1	59.8	6.22×10^{12}	1.04×10^8	0.05
60	H2-A	59.9	1.37×10^{13}	2.29×10^8	0.79
60	H2-C	59.6	1.36×10^{13}	2.28×10^8	0.93
80	H1	79.3	7.60×10^{12}	9.59×10^7	0.08
80	H2-A	80.0	1.52×10^{13}	1.90×10^8	1.45
80	H2-C	79.0	1.47×10^{13}	1.86×10^8	1.56

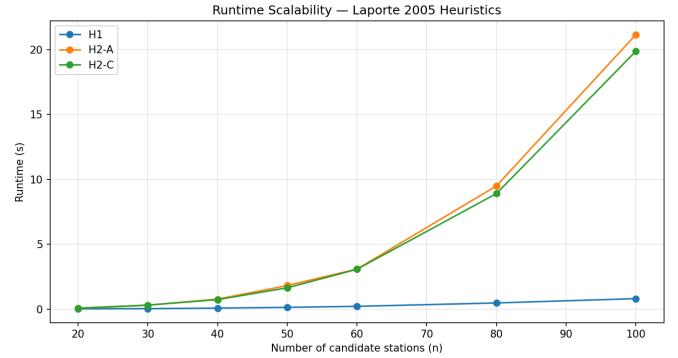


Figure 5. Runtime scaling: H1 as $O(n^3)$, H2 as $O(n^4)$, matching complexity analysis.

finding, though smaller than its 70% margin due to our smaller candidate set (54 vs. 4,774).

D. CLPS Line-Pool Evaluation

We separately evaluate CLPS on the generated candidate line pool. After deduplication and length filtering, the pool contains 400 candidate lines. Their lengths range from 3.9 km to 26.3 km, with an average of 10.8 km; each line contains 4–12 stations, averaging 6.7 stations. This confirms the intended reduction: instead of searching over all possible station sequences, the selectors operate on a compact but diverse set of plausible lines.

Table IV shows three clear behaviors. Greedy-OD and ILP select the same line set in this run, indicating that the greedy marginal OD rule reaches the direct-OD optimum for this pool and constraint setting. Greedy-BCR produces the most compact network, using only 35.4 km of unique track, but sacrifices coverage. GA-pool selects a broader network that covers 50 stations and achieves the highest network OD coverage (96.3%), at the cost of longer track length and slightly larger weighted detour. These results support the motivation for CLPS: once the route space is reduced to a candidate line pool, different selectors can explore coverage-cost trade-offs at very low computational cost.

E. Cross-Method Comparison

Table V shows: (1) **Coverage**: Laporte H2-C achieves the highest OD coverage (56.8%) with 33 stations, but at higher detour ratio (2.19). (2) **Directness**: Ahmed-GA-3L achieves

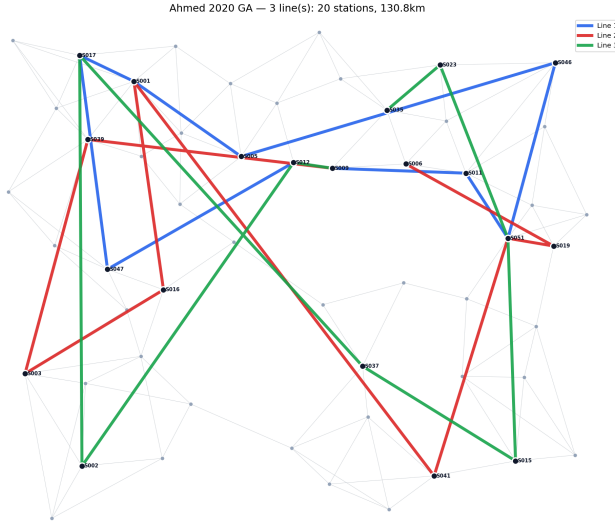


Figure 6. 3-line GA result: hub-based radial topology centered on S009.

Table II
AHMED 2020 GA RESULTS.

Lines	Stations	Length (km)	Cost	Time (s)
1	14	37.0	-5.63×10^{12}	33.9
2	23	85.7	-1.33×10^{13}	57.1
3	20	130.8	-5.81×10^{12}	59.6

the best detour ratio (0.14) but lower OD coverage (26.0%). (3) **Transfers:** Only multi-line methods support transfers. (4) **Efficiency:** CLPS-Greedy-OD runs in 0.01 s with 51.9% OD coverage, showing the computational advantage of reducing route design to line-pool selection. The ILP selector is implemented in the same CLPS framework; in the current run it selected the same line set as CLPS-Greedy-OD, so we omit a duplicate table row.

F. Passenger Flow Analysis (Stage 5 Extension)

We perform all-or-nothing passenger assignment to evaluate operational realism. Table VI reports section-level flow statistics. Laporte-H1 shows highest concentration (top-5 edges carry 56.9% of flow), indicating a linear bottleneck. Laporte-H2A achieves the most balanced distribution (20.3%). Ahmed-GA-2L concentrates flow (49.9%) around hub S009, while Ahmed-GA-3L achieves the lowest peak flow (0.12) as the third line diverts demand. These patterns validate that single-path methods create bottlenecks while multi-line networks distribute load across corridors.

G. Sensitivity Analysis

Spacing sensitivity (500–1,500 m) showed no effect because all corridor edges exceed 1,000 m—a limitation of our sparse layout. GA convergence (Fig. 8) stabilizes within 10–15 generations, confirming the paper’s $P_c = 0.7$ recommendation.

Table III
SIMULTANEOUS VS. INDIVIDUAL OPTIMIZATION (2 LINES).

Method	Cost	Stations	Len. (km)	T (s)
Simultaneous	-1.33×10^{13}	23	85.7	58.3
Individual	-1.13×10^{13}	28	80.1	68.9

Table IV
CLPS SELECTOR RESULTS ON THE 400-LINE CANDIDATE POOL
($K_{\max} = 8$).

Selector	Lines	Sta.	Len. (km)	Pop. Cov.	OD Cov.	Net. Cov.
Greedy-OD	8	42	101.2	90.0%	68.5%	87.9%
Greedy-BCR	8	21	35.4	67.9%	52.1%	69.0%
GA-pool	8	50	120.6	97.9%	67.7%	96.3%
ILP	8	42	101.2	90.0%	68.5%	87.9%

Table V
COMPREHENSIVE COMPARISON OF ALL METHODS ON SHANGHAI DATA.

Method	Stations	Len. (km)	Pop. Cov.	OD Cov.	Detour	Trans.	T (s)
Laporte-H1	18	59.8	48.9%	27.9%	1.68	0.00	0.05
Laporte-H2A	31	59.9	69.5%	54.0%	1.58	0.00	0.77
Laporte-H2C	33	59.6	69.6%	56.8%	2.19	0.00	0.88
Ahmed-GA-1L	14	37.0	44.0%	29.8%	4.43	0.00	33.5
Ahmed-GA-2L	23	85.7	61.9%	42.5%	1.23	0.13	56.3
Ahmed-GA-3L	20	130.8	47.8%	26.0%	0.14	0.18	58.1
CLPS-Greedy-OD	23	—	62.2%	51.9%	1.21	0.08	0.01
CLPS-Greedy-BCR	10	—	35.2%	24.5%	1.19	0.10	0.01
CLPS-GA-pool	32	—	66.0%	34.0%	3.61	0.16	0.90

Table VI
SECTION-LEVEL PASSENGER FLOW STATISTICS.

Method	Max Flow	Mean Flow	Top-5 Share	Edges
Laporte-H1	0.16	0.033	56.9%	28/34
Laporte-H2A	0.14	0.045	20.3%	60/60
Ahmed-GA-2L	0.22	0.027	49.9%	40/52
Ahmed-GA-3L	0.12	0.016	39.4%	43/48

V. CONCLUSION

This project combines reproduction with a new reduced-search line-planning framework on a common Shanghai candidate graph. Key findings: (1) the Laporte reproduction verifies the expected runtime scaling, but our simplified model (no logit mode choice) reverses the paper’s H1-vs-H2 ranking; H2-C achieves the highest OD coverage (56.8%) by packing more stations into one alignment. (2) The Ahmed reproduction produces realistic hub-based multi-line networks; simultaneous optimization outperforms individual optimization by 18%. (3) CLPS provides a computationally light third family of methods: CLPS-Greedy-OD reaches 51.9% OD coverage in 0.01 s by selecting from a precomputed pool of plausible lines. Together, the results show that classical constructive heuristics, direct multi-line GA search, and line-pool selection expose different trade-offs between coverage, directness, transfers, and runtime. Future work: real OD data, logit mode choice, and parallel GA evaluation for larger candidate sets.

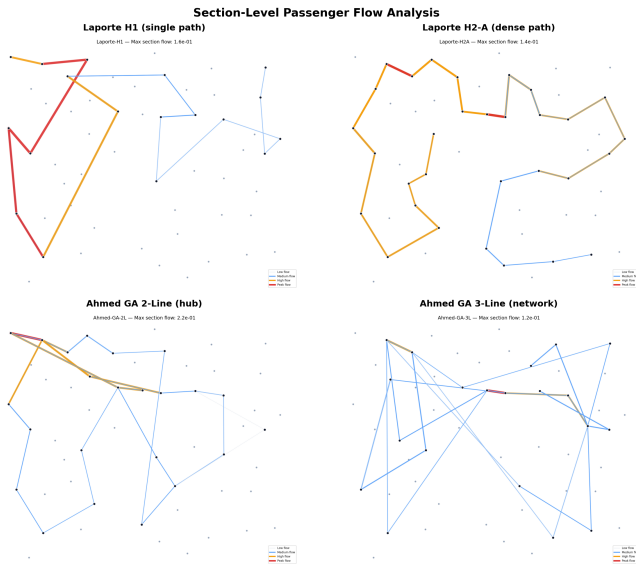


Figure 7. Passenger flow intensity on four networks (gray=low, blue=medium, orange=high, red=peak).

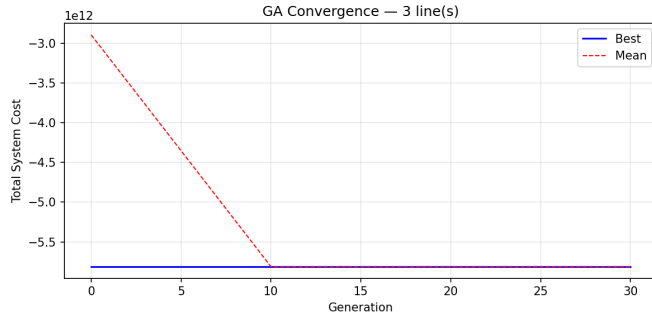


Figure 8. GA convergence for the 3-line scenario.

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